

Product Data API · One-page findings memo

Open Food Facts snapshot · random sample of 5,000 records · seed 42

1. Barcode integrity: 99 of 5,000 sampled records (2.0%) fail the GS1 Mod-10 check-digit rule; 0 barcode groups contain duplicates.

2. Content completeness: average score is 54.3%; 1,161 products (23.22%) score at least 75. *allergens* is most often blank (4,482; 89.6%).

3. Snapshot vs live: 14 of 19 successful comparisons improved, 1 declined and 4 were unchanged (from 20 attempted). This small, valid-barcode sample is directional only; nonresponse and matching/selection bias limit inference.

Recommendation

Prioritize fixing allergens capture and reject invalid GTIN check digits at ingestion, with a correction queue for the 99 affected sample rows. Recheck completeness after remediation.

What would increase confidence

Repeat the comparison on a larger stratified sample across brands, categories and score bands; retry 404/429/timeouts on a controlled schedule, record timestamps, and compare multiple snapshots over time.

Issue report · invalid GS1 check digits

Evidence: `SELECT COUNT(*) FROM products WHERE barcode_valid=0`; result = 99/5,000. Expected: invalid barcodes are flagged and excluded from reliable joins. Actual: these records can fail lookup or be incorrectly linked. Impact: 99 sample records. 90-day cost assumption: 2 minutes of manual triage per affected row once = 3.3 staff hours; errors may also mislead shoppers and catalog teams. Screenshot: `issue-screenshot.svg` in `reports/`.